Chapter 19 Problem Set

Siddharth K

February 2024

1a. This is a linear recurrence who's characteristic polynomial is $P(x) = x^2 - 3x + 2$, and its roots are $-1, -2$. We know that there exist s_1 and s_2 such that $a_n = s_1(-1)^n + s_2(-2)^n$. Since $a_0 = 1$, $s_1 + s_2 = 1$, and since $a_1 = 3$, $-s_1 - 2s_2 = 3$. Solving this system, we get $s_1 = 5$ and $s_2 = -4$. Therefore, $a_n = 5(-1)^n - 4(-2)^n$.

1b. This recurrence's characteristic polynomial is $P(x) = x^2 - 4x + 1$, and its roots are $2 + \sqrt{3}, 2 - \sqrt{3}$. So there exist s_1 and s_2 such that $a_n = s_1(2 + \sqrt{3})^n + s_2(2 - \sqrt{3})^n$. This gives us $s_1 + s_2 = 2$ and $s_1(2+\sqrt{3}) + s_2(2-\sqrt{3}) = 3$. Solving that system, we get $s_1 = \frac{6-\sqrt{3}}{6}$ and $s_2 = \frac{6+\sqrt{3}}{6}$. Therefore, $a_n = \frac{6-\sqrt{3}}{6}(2 + \sqrt{3})^n + \frac{6+\sqrt{3}}{6}(2 - \sqrt{3})^n$.

2. Let a_n represent the number of pairs of rabbits after n months. For $n > 2$, each pair of rabbits that was alive in month $n - 1$ will still be alive, and each pair of rabbits from month $n-2$ would have each produced one pair. Therefore, $a_n = a_{n-1} + a_{n-2}$. Since $a_0 = 1$ and $a_1 = 1$, this sequence is just the Fibonacci sequence, where $a_n = F_{n+1}$ ($F_0 = 0$ and $F_1 = 1$). Therefore, $a_{12} = F_{13} = 233$.

3. Let's consider the ways to fill the rightmost part of the rectangle. If we choose a 1×1 space, there are 5 choices and a_{n-1} for the rest of the rectangle. If we choose a 1×2 space, there are 4 choices and a_{n-2} for the rest of the rectangle. Therefore, the recurrence relation is $a_n = 5a_{n-1} + 4a_{n-2}$. There is one way to fill a 1×0 grid and 5 ways to fill a 1×1 grid, so $a_0 = 1$ and $a_1 = 5$.

Now, we have a linear recurrence relation and its initial conditions. Its characteristic polynomial is $P(x) = x^2 - 5x - 4$ and its roots are $\frac{5 \pm \sqrt{41}}{2}$. That means there exist s_1 and s_2 such that $a_n = s_1 \left(\frac{5+\sqrt{41}}{2}\right)^n + s_2 \left(\frac{5-\sqrt{41}}{2}\right)^n$. Using the initial conditions, we get the system of equations:

$$s_1 + s_2 = 1,$$

$$s_1 \left(\frac{5 + \sqrt{41}}{2}\right) + s_2 \left(\frac{5 - \sqrt{41}}{2}\right) = 5.$$

Solving it, we get $s_1 = \frac{41+5\sqrt{41}}{84}$ and $s_2 = \frac{41-5\sqrt{41}}{84}$. Therefore, $a_n = \left(\frac{41+5\sqrt{41}}{84}\right)\left(\frac{5+\sqrt{41}}{2}\right)^n + \left(\frac{41-5\sqrt{41}}{84}\right)\left(\frac{5-\sqrt{41}}{2}\right)^n$.

4. The recurrence's solution is $a_n = \lfloor \log_2(n) \rfloor + 7$, and we can prove this using induction. When $n = 1$, $a_n = 7 = \lfloor \log_2(1) \rfloor + 7$.

Now, let's assume $a_n = \lfloor \log_2(n) \rfloor + 7$. By the given recurrence, we see that

$$a_{2n} = a_n + 1 = \lfloor \log_2(n) \rfloor + 8 = \lfloor \log_2(2n/2) \rfloor + 8$$

$$= \lfloor \log_2(2n) - \log_2(1/2) \rfloor + 8 = \lfloor \log_2(2n) - 1 \rfloor + 8 = \lfloor \log_2(2n) \rfloor + 7,$$

as desired.

5. Let's consider the position of element 1 in a derangement of $[n]$. That element must move to the position of element k where $2 \leq k \leq n$. Let's assume k doesn't move to the first spot. Then, we need to find a derangement of the remaining $n-1$ elements. Since there are $n-1$ ways to choose k, there are $(n-1)d(n-1)$ ways to create a derangement this way.

Now, let's assume 1 and k switch spots. Then we are left with $n-2$ elements to find a derangement of, so the total number of ways to form a derangement of $[n]$ this way is $(n-1)d(n-2)$. Adding both cases, we get $d(n) = (n-1)(d(n-1) + d(n-2))$.

7. In the process of flipping coins, Bob's score must have been either $n-2$ or $n-1$. If it was $n-1$, the probability that his score becomes n is $\frac{1}{2}$. If his score was $n-2$, the probability that his score becomes n without reaching $n-1$ is $\frac{1}{2}$ (if it does reach $n-1$ then it counts as the earlier case). So $p_n = \frac{1}{2}p_{n-1} + \frac{1}{2}p_{n-2}$. Also, $p_0 = 1$ since the score is always 0 at the start, and $p_1 = \frac{1}{2}$ since, after the first flip, his score either reaches 1 or 2 with equal probability.

Now, we have a linear recurrence and two initial conditions. The recurrence's characteristic polynomial is $P(x) = x^2 - \frac{1}{2}x - \frac{1}{2}$, and its roots are $\frac{1}{2}$ and 1. That means there exist s_1 and s_2 such that $p_n = s_1(\frac{1}{2})^n + s_2(1)^n = \frac{s_1}{2^n} + s_2$. We know that $p_0 = 1$, so $s_1 + s_2 = 1$. Also, $p_1 = \frac{1}{2}$, so $\frac{s_1}{2} + s_2 = \frac{1}{2}$. Solving the system, we get $s_1 = 1$ and $s_2 = 0$. Therefore, $p_n = \frac{1}{2^n}$.

8. If $a_n + b_n\sqrt{3} = (2+\sqrt{3})^{2n-1}$, then

$$a_{n+1} + b_{n+1}\sqrt{3} = (2+\sqrt{3})^{2n+1} = (2+\sqrt{3})^2(a_n + b_n\sqrt{3}) = (7+4\sqrt{3})(a_n + b_n\sqrt{3})$$

$$= (7a_n + 12b_n) + (4a_n + 6b_n)\sqrt{3}.$$

This gives us $a_{n+1} = 7a_n + 12b_n$. Since $a_n + b_n\sqrt{3} = (2+\sqrt{3})^{2n-1}$, we know that $b_n = \frac{\sqrt{3}}{3}((2+\sqrt{3})^{2n-1} - a_n)$. Substituting that into the equation, we get

$a_{n+1} = (7 - 4\sqrt{3})a_n + 4\sqrt{3}(2 + \sqrt{3})^{2n-1}$. Also, $a_1 = 2$. I tried proving that $a_n - 1$ is a perfect square by induction, but it didn't work.

9. The closed formula for the Josephus problem is $J(n) = 2(n - 2^{\lfloor \log_2 n \rfloor}) + 1$. The recurrence is $J(2n) = 2J(n) - 1$ and $J(2n + 1) = 2J(n) + 1$, and the initial condition is $J(1) = 1$. That matches the closed form because $J(1) = 2(1 - 2^0) + 1 = 1$.

Let's assume that $J(n) = 2(n - 2^{\lfloor \log_2 n \rfloor}) + 1$. We need to prove that $J(2n) = 2(2n - 2^{\lfloor \log_2 (2n) \rfloor}) + 1$. By the recurrence,

$$J(2n) = 2J(n) - 1 = 2(2(n - 2^{\lfloor \log_2 n \rfloor}) + 1) - 1 = 2(2n - 2^{\lfloor \log_2 n \rfloor + 1}) + 1.$$

Also, we have $\lfloor \log_2 n \rfloor + 1 = \lfloor \log_2 n + 1 \rfloor = \lfloor \log_2 n + \log_2 2 \rfloor = \lfloor \log_2 (2n) \rfloor$. Therefore, $J(2n) = 2(2n - 2^{\lfloor \log_2 (2n) \rfloor}) + 1$, as desired.

Now, we need to prove that $J(2n + 1) = 2((2n + 1) - 2^{\lfloor \log_2 (2n+1) \rfloor}) + 1$. By the recurrence,

$$J(2n + 1) = 2J(n) + 1 = 2(2n - 2^{\lfloor \log_2 (2n) \rfloor} + 1) + 1 = 2((2n + 1) - 2^{\lfloor \log_2 (2n) \rfloor}).$$

Also, $\lfloor \log_2 (2n) \rfloor \neq \lfloor \log_2 (2n + 1) \rfloor$ if and only if $2n$ is one less than a power of 2, which is impossible since it is even. Therefore, $\lfloor \log_2 (2n) \rfloor = \lfloor \log_2 (2n + 1) \rfloor$ and $J(2n + 1) = 2((2n + 1) - 2^{\lfloor \log_2 (2n+1) \rfloor})$, as desired.